

Python Built-in Exceptions Cheat Sheet

This cheat sheet provides a quick reference to common built-in exceptions in Python. Exceptions are errors that occur during program execution, and understanding them can help in debugging and writing more resilient code.

Exception

Base class for most built-in exceptions. All custom exceptions should inherit from this.

ArithmeticError

Base class for numeric calculation errors.

ZeroDivisionError

Raised when dividing by zero.

OverflowError

Raised when a numeric calculation exceeds the representable range.

AssertionError

Raised when an assert statement fails.

AttributeError

Raised when an invalid attribute reference is made.

EOFError

Raised when input() hits end-of-file (no more data).

ImportError

Raised when an import fails.

ModuleNotFoundError

Subclass of ImportError; raised when a module can't be found.

IndexError

Raised when a sequence index is out of range.

KeyError

Raised when a dictionary key is not found.

KeyboardInterrupt

Raised when the user presses Ctrl+C.

MemoryError

Raised when an operation runs out of memory.

NameError

Raised when a variable is not defined.

UnboundLocalError

Subclass of NameError; raised when referencing a local variable before assignment.

OSError

Base class for operating system errors.

FileNotFoundError

Raised when a file or directory is not found.

PermissionError

Raised when an operation lacks proper permissions.

IsADirectoryError

Raised when a file operation is attempted on a directory.

NotADirectoryError

Raised when a directory operation is attempted on something that isn't a directory.

RuntimeError

Raised when an unspecified runtime error occurs.

RecursionError

Raised when maximum recursion depth is exceeded.

StopIteration

Raised to signal no more items from an iterator.

StopAsyncIteration

Raised to signal no more items from an async iterator.

SystemExit

Raised by `sys.exit()` to exit the program.

TypeError

Raised when an operation is applied to an object of inappropriate type.

ValueError

Raised when an operation receives an invalid value of the correct type.

UnicodeError

Base class for Unicode-related errors.

UnicodeEncodeError

Raised when a string can't be encoded.

UnicodeDecodeError

Raised when a byte sequence can't be decoded.

UnicodeTranslateError

Raised when Unicode translation fails.